

Teacher Teaching Script

MHF4U — Chapter 2 — Long Division & Synthetic Division

Aligned page-by-page to the finalized 50-page student HTML

How to use this script

Treat the wording as natural speaking guidance, not something that must be read exactly. Pause on the student checkpoints. Let the student calculate on paper before you explain the answer. Keep the lesson centered on signs, missing powers, quotient degree, remainders, and choosing the appropriate method.

Page 1 — Long Division & Synthetic Division

Purpose: Set the purpose and establish what the student will do today.

Teacher script: "Today we are focusing on polynomial division. We are going to learn two actual division methods—long division and synthetic division—and one shortcut for finding a remainder when we do not need the whole division. I do not want you to memorize a page of steps. I want you to notice the pattern underneath them. By the end, you should be able to look at a question and decide what information you need and which method makes the most sense.

Emphasize: Listen for the idea that the method depends on the question, not on habit.

Transition: We start with the structure of division itself.

Page 2 — Division is a relationship

Purpose: Connect ordinary division to polynomial division.

Teacher script: "Let us start with something familiar. 107 divided by 4 can be written as 4 times 26 plus 3. The important part is not the numbers. It is the relationship: dividend equals divisor times quotient plus remainder. We use exactly the same relationship with polynomials: $P(x)$ equals $D(x)Q(x)$ plus $R(x)$.

Ask: What plays the role of the dividend, divisor, quotient, and remainder?

Expected response: $P(x)$ is the dividend, $D(x)$ is the divisor, $Q(x)$ is the quotient, and $R(x)$ is the remainder.

If correct: Exactly. That relationship is our check for every division problem.

If not correct: If the labels are mixed up, point back to the numerical example and match each part before continuing.

Emphasize: A division answer is not just a quotient; the remainder has a place in the relationship.

Transition: Now we will recall how the division process itself works.

Page 3 — The four moves

Purpose: Rebuild the familiar long-division rhythm before using polynomials.

Teacher script: "Before we put x into the problem, tell me what you do in long division. We have four moves: divide, multiply, subtract, and repeat. I want you to see these as a cycle. In polynomial long division, the same cycle happens; the only difference is that our pieces are algebraic terms instead of digits.

Ask: Which move comes immediately after you choose the next quotient term?

Expected response: Multiply.

If correct: Right. And after multiplying, we subtract before bringing the next part into view.

If not correct: If they say bring down, ask: "What must we do with the quotient term we just chose first?"

Emphasize: Keep the four moves in the same order throughout the lesson.

Transition: Let us do one quick number example so the pattern is active in your head.

Page 4 — Quick number check

Purpose: Wake up the number-division pattern.

Teacher script: "Try this one without overthinking it: 753 divided by 22. I do not need a speech. Just get the quotient and remainder. Work it out on paper.

Ask: What is the remainder?

Expected response: 5

If correct: Good. Keep 34 and 5 in mind because we are about to transfer the same process to polynomial terms.

If not correct: Ask them to check $22 \times 34 = 748$ and $753 - 748 = 5$.

Emphasize: The point is the process, not speed.

Transition: Now replace the numbers with powers of x .

Page 5 — From number division to polynomial division

Purpose: Bridge the numerical pattern to algebraic division.

Teacher script: "Look at this polynomial: $x^3 + 3x^2 - 2x + 5$ divided by $x + 1$. The first thing I want you to notice is that we compare the leading terms, just like 753 and 22 led the number division. x^3 divided by x gives x^2 . That becomes our first quotient term.

Ask: Why do we start with x^3 divided by x ?

Expected response: Because the leading terms determine the first term of the quotient.

If correct: Exactly. We are matching degrees as well as coefficients.

If not correct: Bring their attention back to the highest-degree terms only.

Emphasize: Start with the first terms of the dividend and divisor.

Transition: Before dividing, we need the polynomial written properly.

Page 6 — Get the polynomial ready

Purpose: Teach descending powers and zero coefficients.

Teacher script: "Before we divide, check the polynomial. $x^4 - 4x^2 - 2x + 3$ is missing the x^3 term. We write it as $x^4 + 0x^3 - 4x^2 - 2x + 3$. That zero is not decoration. It holds the place of the missing power so every term lines up correctly. This matters in both long division and synthetic division.

Ask: Why do we put a zero before the $-4x^2$ term?

Expected response: Because the x^3 term is missing, and the zero keeps the powers aligned.

If correct: Yes. Missing powers must still occupy a place.

If not correct: Ask what happens to the coefficient row if the zero is skipped.

Emphasize: Descending order and zero placeholders are non-negotiable.

Transition: Now let us do the full long division.

Page 7 — First worked polynomial division

Purpose: Model a complete long-division example.

Teacher script: "Let us work exactly one complete example together: divide $3x^3 - 5x^2 - 7x - 1$ by $x - 3$. First terms: $3x^3$ divided by x gives $3x^2$. Multiply $3x^2$ by $x - 3$. We get $3x^3 - 9x^2$. Subtract. That leaves $4x^2 - 7x$. Next, $4x^2$ divided by x gives $4x$. Multiply, subtract, and continue. We get $5x - 1$. Then $5x$ divided by x gives 5 . Multiply and subtract. The remainder is 14 .

Ask: After the first subtraction, what is the new leading term?

Expected response: $4x^2$.

If correct: Good. That is the term we use for the next quotient term.

If not correct: Recheck the subtraction: $(-5x^2) - (-9x^2) = 4x^2$.

Emphasize: Speak through the sign changes slowly.

Transition: Next we will slow the process down and explain why each line exists.

Page 8 — What each step is doing

Purpose: Explain the logic behind the long-division lines.

Teacher script: "Do not just copy the layout. The first quotient term is there because $3x^3$ divided by x gives $3x^2$. Then we multiply that entire quotient term by the divisor so the leading term cancels. After subtraction we have a new expression to work with. We repeat until the remainder has smaller degree than the divisor.

Ask: When do we stop?

Expected response: When the degree of the remainder is less than the degree of the divisor.

If correct: Exactly. With a linear divisor of degree 1, we stop at a constant remainder.

If not correct: Ask them to compare the degree of the current remainder with the divisor.

Emphasize: The stopping rule is about degree, not about whether the remainder is zero.

Transition: Now let us write the same result in the forms the course expects.

Page 9 — Quotient form vs. division statement

Purpose: Teach the required answer formats and checking identity.

Teacher script: "There are two useful ways to write the same result. Quotient form is $3x^2 + 4x + 5 + 14 \text{ over } x - 3$. The division statement is the cleaner checking form: $3x^3 - 5x^2 - 7x - 1 = (x - 3)(3x^2 + 4x + 5) + 14$. In this course, you should be comfortable with both.

Ask: What is the remainder in the division statement?

Expected response: 14.

If correct: Yes. And notice the remainder is added after the divisor times quotient.

If not correct: If they choose $14/(x-3)$, explain that belongs in quotient form, not in the division identity.

Emphasize: Do not lose the remainder or attach it to the wrong part.

Transition: Let us make sure we can read a final division line correctly.

Page 10 — Checkpoint: read the remainder

Purpose: Check whether the student can identify the final synthetic/long-division output.

Teacher script: "Look at the final line. If the last value is 14 and the divisor has degree 1, what is that 14? Choose one before we discuss it.

Ask: What does 14 represent?

Expected response: The remainder.

If correct: Correct. The last number is the remainder.

If not correct: Ask: "Which part comes after the quotient coefficients in the division identity?"

Emphasize: Have the student say the words "quotient coefficients" and "remainder" rather than only pointing.

Transition: Now we deal with divisors whose leading coefficient is not 1.

Page 11 — Non-monic divisors in long division

Purpose: Prepare the student for divisors like $3x+1$.

Teacher script: "The review includes divisors such as $3x + 1$. That changes the first quotient step because $12x^3$ divided by $3x$ is $4x^2$ —not $12x^2$. Let us keep the example in front of us: $12x^3 - 2x^2 + x - 11$ divided by $3x + 1$ gives quotient $4x^2 - 2x + 1$ and remainder -12 .

Ask: What is $12x^3$ divided by $3x$?

Expected response: $4x^2$.

If correct: Exactly. Divide both coefficient and power.

If not correct: If they say $4x^3$, ask what happens to x^3 / x .

Emphasize: The coefficient of x in the divisor matters immediately.

Transition: Let us check the first quotient term before doing the rest.

Page 12 — Checkpoint: the first quotient term

Purpose: Check division of leading terms with a non-monic divisor.

Teacher script: "Just the first move: $12x^3 - 2x^2 + x - 11$ divided by $3x + 1$. What is the first quotient term? Take a moment and choose.

Ask: What is the first quotient term?

Expected response: $4x^2$.

If correct: Yes. $12/3$ gives 4 and x^3/x gives x^2 .

If not correct: Ask them to say both parts: coefficient and variable power.

Emphasize: This is the same idea as the previous page, but the leading coefficient cannot be ignored.

Transition: After division, we need to know how to prove our result is correct.

Page 13 — Why verification matters

Purpose: Teach the checking identity as an error detector.

Teacher script: "One habit I want you to keep is checking. For $x^3 + 9x^2 - 5x + 3$ divided by $x - 2$, the quotient is $x^2 + 11x + 17$ and the remainder is 37. Rebuild the dividend: $(x - 2)(x^2 + 11x + 17) + 37$. If that expands back to the original polynomial, our division is consistent.

Ask: What should the right side simplify to?

Expected response: The original dividend, $x^3 + 9x^2 - 5x + 3$.

If correct: Exactly. The check is a reconstruction of the original polynomial.

If not correct: Ask which part they should inspect if the expansion does not match: multiplication, signs, or remainder.

Emphasize: Verification is especially useful when the numbers get larger.

Transition: Now let us catch a realistic error.

Page 14 — Checkpoint: find the mistake

Purpose: Use a checking statement as error analysis.

Teacher script: "Suppose a student has $Q(x) = x^2 + 11x + 17$ and $R = 37$. Which final checking statement is correct? Read all four carefully.

Ask: Which statement is correct?

Expected response: $(x - 2)(x^2 + 11x + 17) + 37$.

If correct: Yes. That is exactly D times Q plus R.

If not correct: Ask what the division relationship from page 2 says.

Emphasize: Make them connect the general identity to a concrete problem.

Transition: Now we use a shortcut when only the remainder is needed.

Page 15 — Remainder Theorem

Purpose: Introduce the theorem and when to use it.

Teacher script: "Suppose the question asks only for the remainder when $P(x)$ is divided by $x - b$. We do not need the whole division. The remainder is $P(b)$. So the first thing we do is find the zero of the divisor. This is the Remainder Theorem, and it is one of the quickest tools in this section.

Ask: If the divisor is $x - 5$, what value goes into P?

Expected response: 5.

If correct: Right. We use the zero of the divisor.

If not correct: Ask them to solve $x - 5 = 0$ before using the theorem.

Emphasize: Always identify the divisor zero before substituting.

Transition: Let us do the supplied review-style example.

Page 16 — Remainder Theorem example

Purpose: Model a complete remainder-only question.

Teacher script: "Use the supplied review example: $P(x) = x^4 - 4x^2 - 2x + 3$, divided by $x + 1$. First rewrite $x + 1$ as $x - (-1)$. So we use $P(-1)$. Then substitute carefully: $1 - 4 + 2 + 3 = 2$. The remainder is 2. Notice how much shorter this is than long division.

Ask: Why do we use -1, not +1?

Expected response: Because $x + 1 = x - (-1)$, so its zero is -1.

If correct: Exactly. The sign comes from the zero of the divisor.

If not correct: Have them solve $x + 1 = 0$ aloud.

Emphasize: Sign handling is the first thing to check.

Transition: Let us isolate that sign issue with a quick checkpoint.

Page 17 — Checkpoint: sign first

Purpose: Check sign interpretation for $x+3$.

Teacher script: "For the divisor $x + 3$, stop before calculating anything. What value should be substituted into $P(x)$?"

Ask: What value?

Expected response: -3.

If correct: Correct. $x + 3 = 0$ gives $x = -3$.

If not correct: Ask them to write $x + 3$ as $x - (-3)$.

Emphasize: Do not let them memorize "copy the sign"; make them solve the zero.

Transition: The same theorem also works when the coefficient of x is not 1.

Page 18 — The non-monic Remainder Theorem case

Purpose: Teach the $ax-b$ form from the supplied lesson.

Teacher script: "The course also gives the more general case. If the divisor is $ax - b$, the zero is b over a , so the remainder is $P(b/a)$. For $2x - 3$, solve $2x - 3 = 0$ and get $x = 3/2$. That is the value we substitute.

Ask: For $2x - 3$, what is the divisor zero?

Expected response: $3/2$.

If correct: Exactly. This is the value used in the theorem.

If not correct: Ask them to solve the linear equation rather than guess.

Emphasize: The quotient is not involved when the question asks only for the remainder.

Transition: Let us calculate one so the fraction does not feel abstract.

Page 19 — Worked non-monic remainder

Purpose: Model the $2x-3$ Remainder Theorem calculation.

Teacher script: "Take $P(x) = 2x^3 + x^2 - 3x - 6$ divided by $2x - 3$. We use $x = 3/2$. Substitution gives $P(3/2) = -3/2$. That is the remainder. The supplied teacher material also checks this result by long division, which is useful because it shows the theorem and division agree.

Ask: What is the remainder?

Expected response: $-3/2$.

If correct: Yes.

If not correct: If they get a different value, recompute the powers of $3/2$ one line at a time.

Emphasize: Fractions are normal here. Do not turn a fraction into a decimal unless needed.

Transition: Now we use remainder information to find an unknown coefficient.

Page 20 — Finding an unknown coefficient

Purpose: Introduce unknown-k remainder problems.

Teacher script: "This type of question gives us a polynomial with an unknown coefficient and tells us the remainder. For $P(x) = x^4 + kx^3 - 3x - 5$ divided by $x - 3$, the remainder is -10 . The important move is not division. It is to translate the remainder statement into $P(3) = -10$.

Ask: What equation should we write before solving for k ?

Expected response: $P(3) = -10$.

If correct: Exactly. The theorem gives the equation we need.

If not correct: Ask: "What value belongs in $P(x)$ because the divisor is $x-3$?"

Emphasize: Make the student state the theorem before substituting.

Transition: Now solve the algebra carefully.

Page 21 — Solve for k

Purpose: Model the algebra in the unknown-coefficient problem.

Teacher script: "Substitute 3 into the polynomial: $81 + 27k - 9 - 5 = -10$. Combine constants: $27k + 67 = -10$. Then $27k = -77$, so $k = -77/27$. The fraction is the exact answer.

Ask: What is k ?

Expected response: $-77/27$.

If correct: Correct. Check by substituting it back into the original remainder equation if you want a quick confirmation.

If not correct: If they make an arithmetic error, have them recalculate the constants first, then isolate k .

Emphasize: Keep the algebra separate from the theorem: first translate, then solve.

Transition: Now we make a choice: when do we even need to divide?

Page 22 — Checkpoint: choose the tool

Purpose: Teach method selection for remainder-only problems.

Teacher script: "You only need the remainder when $P(x)$ is divided by $x - 5$. Which is the most direct first move?

Ask: Which method?

Expected response: Evaluate $P(5)$ using the Remainder Theorem.

If correct: Exactly. No full division is necessary when only the remainder is requested for $x-b$.

If not correct: Ask what extra work long division would create.

Emphasize: The fastest method is the one that answers the exact question asked.

Transition: Now move from the shortcut back to full division—but with a shorter layout.

Page 23 — Synthetic Division: what changes?

Purpose: Introduce synthetic division as compressed polynomial division.

Teacher script: "Synthetic division does not create a different mathematical result. It compresses the arithmetic of polynomial division into a coefficient row. For $x - b$, place b outside, write the coefficients of the dividend, bring the first coefficient down, multiply, add, and repeat. The final number is the remainder.

Ask: What information from the full polynomial do we keep?

Expected response: The coefficients, in descending powers, plus the divisor zero.

If correct: Right. The variable expressions are rebuilt afterward.

If not correct: If they mention only the numbers, remind them that the coefficients must include zeros for missing powers.

Emphasize: Same arithmetic idea, less writing.

Transition: Let us run the complete table one move at a time.

Page 24 — Synthetic division, one move at a time

Purpose: Model the synthetic algorithm explicitly.

Teacher script: "For $3x^3 - 5x^2 - 7x - 1$ divided by $x - 3$, put 3 outside and the coefficients 3, -5, -7, -1 inside. Bring down 3. Multiply 3 by 3 to get 9. Add to -5 to get 4. Multiply 4 by 3 to get 12. Add to -7 to get 5. Multiply 5 by 3 to get 15. Add to -1 to get 14. Bottom row: 3, 4, 5, 14.

Ask: What is the first multiplication?

Expected response: $3 \times 3 = 9$.

If correct: Yes. Then that 9 is added to the next coefficient.

If not correct: If they skip the bring-down step, stop and make that the first step again.

Emphasize: Keep saying "multiply, then add" so the rhythm becomes automatic.

Transition: Now turn the bottom row back into an algebraic quotient.

Page 25 — How to rebuild the quotient

Purpose: Teach degree and coefficient interpretation of synthetic output.

Teacher script: "The bottom row is not the final polynomial yet. The first three values, 3, 4, and 5, are quotient coefficients. Because the dividend is degree 3, the quotient is degree 2. So the quotient is $3x^2 + 4x + 5$. The last number, 14, is the remainder.

Ask: Why does 3 become the coefficient of x^2 ?

Expected response: Because dividing degree 3 by degree 1 gives a degree-2 quotient.

If correct: Exactly. The quotient starts one degree lower than the dividend.

If not correct: Ask them what would happen for a degree-4 dividend.

Emphasize: Read the bottom row by degrees, not just from left to right.

Transition: Missing powers are where students often lose the table.

Page 26 — Missing coefficients in synthetic division

Purpose: Practice setting up zeros correctly.

Teacher script: "Take $x^3 - 7x - 6$. There is no x^2 term. The coefficient row must be 1, 0, -7, -6. That zero is part of the problem. If you leave it out, every later value shifts into the wrong degree.

Ask: What is the coefficient row?

Expected response: 1, 0, -7, -6.

If correct: Correct.

If not correct: Ask them to say the polynomial in full: $x^3 + 0x^2 - 7x - 6$.

Emphasize: Zero coefficients are structural, not optional.

Transition: Let us test whether you can insert the zero yourself.

Page 27 — Checkpoint: insert the zero

Purpose: Check missing-power setup.

Teacher script: "Set up $x^3 - 2x + 1$ for synthetic division. Which coefficient row is correct?"

Ask: Which row?

Expected response: 1, 0, -2, 1.

If correct: Exactly.

If not correct: Ask which power is missing. The missing x^2 term supplies the zero.

Emphasize: Have the student identify the missing power before choosing.

Transition: Now handle $x+3$, where the sign causes trouble.

Page 28 — When the divisor is $x+3$

Purpose: Teach the sign rule by solving for the zero.

Teacher script: "For $x + 3$, synthetic division uses -3 , not $+3$. Do not memorize that as a random sign rule. Solve $x + 3 = 0$. The zero is -3 . That is the number that goes outside the synthetic table.

Ask: What goes outside the table?

Expected response: -3 .

If correct: Correct.

If not correct: If they choose $+3$, ask them to solve the divisor equation again.

Emphasize: Always find the zero of the divisor first.

Transition: Let us use $x+2$ in a full example.

Page 29 — Synthetic division with $x+2$

Purpose: Model a complete $x+b$ synthetic example.

Teacher script: "Divide $x^3 + 2x^2 - 6x + 1$ by $x + 2$. The synthetic number is -2 . Coefficients are $1, 2, -6, 1$. Bring down 1 . Multiply by -2 to get -2 ; add to 2 gives 0 . Multiply 0 by -2 gives 0 ; add to -6 gives -6 . Multiply -6 by -2 gives 12 ; add to 1 gives 13 . So the quotient is $x^2 - 6$ and the remainder is 13 .

Ask: What is the remainder?

Expected response: 13 .

If correct: Yes.

If not correct: If they get a different number, check the sign of the synthetic number first.

Emphasize: Notice how the middle quotient coefficient can be zero; that is okay.

Transition: Next comes the more advanced non-monic synthetic case.

Page 30 — Non-monic divisors in synthetic division

Purpose: Explain how the source handles $ax-b$ in synthetic division.

Teacher script: "The source allows synthetic division with a divisor $ax - b$, but there is an adjustment. For $2x + 3$, factor out the coefficient of x : $2(x + 3/2)$. The zero is $-3/2$. We can use $-3/2$ in the synthetic table, then divide the quotient coefficients by 2 . The lesson also notes that long division may be easier for these divisors.

Ask: What is the synthetic number for $2x-3$?

Expected response: $3/2$.

If correct: Right. Solve $2x - 3 = 0$.

If not correct: If they say 3 , ask what happens when you solve the actual divisor equation.

Emphasize: This is the exact point where method choice matters.

Transition: Let us work the supplied non-monic example completely.

Page 31 — Worked non-monic synthetic example

Purpose: Model the source's $2x+3$ example and the quotient adjustment.

Teacher script: "Use the source example: $6x^3 + 5x^2 - 16x - 15$ divided by $2x + 3$. The zero is $-3/2$. The coefficient row is 6, 5, -16, -15. Synthetic work gives 6, -4, -10, 0. These first three numbers correspond to $6x^2 - 4x - 10$ for division by $x + 3/2$. Because the original divisor is $2x + 3$, divide the quotient by 2: $Q(x) = 3x^2 - 2x - 5$. Remainder 0.

Ask: What does the zero remainder tell us?

Expected response: The division is exact; $2x+3$ is a factor.

If correct: Exactly. This will connect to the Factor Theorem later.

If not correct: If they focus on the factor statement, keep the lesson on division and note the connection only.

Emphasize: This case is more error-prone; check the divisor rewrite before the table.

Transition: Let us test the synthetic number directly.

Page 32 — Checkpoint: non-monic divisor

Purpose: Check the zero used in the synthetic setup.

Teacher script: "For divisor $2x - 3$, what value belongs outside the synthetic table? Choose carefully.

Ask: Which value?

Expected response: $3/2$.

If correct: Correct.

If not correct: Ask them to solve $2x - 3 = 0$.

Emphasize: This is a zero-of-the-divisor question, not a coefficient question.

Transition: Now let us compare the three methods we have learned.

Page 33 — Long division or synthetic division?

Purpose: Teach selection based on the question and divisor.

Teacher script: "Let us compare the tools. If I only need a remainder for $x - b$, the Remainder Theorem is usually the cleanest. If I need the full quotient and the divisor is $x - b$, synthetic division is usually efficient. If the divisor is $ax - b$, long division is often easier to read, especially if the arithmetic is already comfortable. If the divisor is quadratic, use long division rather than forcing synthetic division.

Ask: What would you choose for a full quotient divided by $x-6$?

Expected response: Synthetic division.

If correct: Yes.

If not correct: Ask what makes the divisor fit synthetic division.

Emphasize: Method choice should come from the form of the divisor and what the question asks.

Transition: Let us test that choice without doing the arithmetic.

Page 34 — Checkpoint: choose the method

Purpose: Check method choice for a remainder-only question.

Teacher script: "Question: Find the remainder when $P(x)$ is divided by $x + 4$. Which method is the most direct?"

Ask: Which method?

Expected response: Remainder Theorem.

If correct: Exactly. Use the zero -4 and evaluate $P(-4)$.

If not correct: If they say synthetic, acknowledge that it can produce the remainder but is not the most direct here.

Emphasize: The question asks only for a remainder.

Transition: Now choose a method for a full quotient.

Page 35 — Checkpoint: choose synthetic division

Purpose: Check method choice for a full quotient.

Teacher script: "Question: You need the full quotient when dividing by $x - 6$. Which method is a natural first choice?"

Ask: Which method?

Expected response: Synthetic division.

If correct: Correct. It is designed for this divisor form.

If not correct: Ask what makes $x-6$ fit the synthetic setup.

Emphasize: Be clear that "natural first choice" does not mean the only mathematically possible method.

Transition: Now use actual source review questions.

Page 36 — Source review: three remainders

Purpose: Practice the same theorem on three divisors to make the pattern visible.

Teacher script: "The workbook asks for remainders of $P(x) = 2x^3 + 7x^2 - 8x + 3$ divided by $x+1$, $x-2$, and $x+3$. I want the student to notice that the polynomial does not change. Only the divisor zero changes. For $x+1$ use -1. For $x-2$ use 2. For $x+3$ use -3. Do one together, then let the student do the other two.

Ask: For $x+3$, which value goes into P ?

Expected response: -3.

If correct: Yes.

If not correct: Return to solving the divisor equal to zero.

Emphasize: The repeated structure is the point: same theorem, new zero.

Transition: Now switch back to a full quotient question.

Page 37 — Source review: a full quotient

Purpose: Practice a missing-term full division from the review.

Teacher script: "Use the review problem: $(x^4 - 4x^2 - 2x + 3)$ divided by $x - 2$. Before calculating, ask what is missing. There is no x^3 term, so use coefficient 0. The review result is $Q(x) = x^3 + 2x^2 + 2x - 2$ and $R = -1$. Let the student set it up, then compare their work with the expected result.

Ask: What coefficient represents the missing x^3 term?

Expected response: 0.

If correct: Correct.

If not correct: If they want to jump into arithmetic, stop and fix the setup first.

Emphasize: Setup errors are easier to prevent than to repair later.

Transition: Now use a non-monic full division problem.

Page 38 — Source review: a non-monic full division

Purpose: Reinforce long division with a non-monic divisor.

Teacher script: "Use the review problem $(12x^3 - 2x^2 + x - 11)$ divided by $(3x + 1)$. The expected result is $Q(x) = 4x^2 - 2x + 1$, $R = -12$. Ask the student to build the division and then write the checking statement. This is a good place to compare it with synthetic division and ask which feels clearer.

Ask: What is the verification statement?

Expected response: $12x^3 - 2x^2 + x - 11 = (3x+1)(4x^2 - 2x + 1) - 12$.

If correct: Exactly.

If not correct: Check whether they put the remainder on the correct side and keep its negative sign.

Emphasize: The remainder can be negative; that is completely valid.

Transition: Finish with the workbook application where division has a purpose.

Page 39 — Source review: box dimensions

Purpose: Use polynomial division in a meaningful application.

Teacher script: "Here is the box problem from the workbook. The volume is $2x^3 + 17x^2 + 38x + 15$ and the height is $x + 5$. Ask: if volume equals height times base area, what operation gives the base area? We divide

the volume polynomial by $x+5$. That gives $2x^2 + 7x + 3$, which factors as $(x+3)(2x+1)$. So possible dimensions are $x+5$, $x+3$, and $2x+1$.

Ask: Why do we divide the volume polynomial by the known height?

Expected response: To find the base-area expression.

If correct: Yes. Volume = height $\times$ base area, so base area = volume / height.

If not correct: Ask what the unknown factor represents in the volume model.

Emphasize: Keep the algebra connected to the geometry: the quotient has a physical meaning.

Transition: Now check that interpretation directly.

Page 40 — Checkpoint: application reasoning

Purpose: Check whether the student understands why the division is being used.

Teacher script: "One question before we move on: why are we dividing the volume by the known height in the box problem?"

Ask: Why divide?

Expected response: To find the base-area expression.

If correct: Exactly.

If not correct: Ask what formula connects volume, height, and base area.

Emphasize: The operation should make sense before the arithmetic begins.

Transition: Let us deliberately catch the three most common mistakes.

Page 41 — Error analysis: three common traps

Purpose: Make the student recognize the three most likely errors.

Teacher script: "Let us make the common mistakes visible. Trap one: using $+3$ for $x+3$ in synthetic division. Trap two: skipping a zero for a missing power. Trap three: using 3 instead of $\frac{3}{2}$ for $2x-3$. These are setup errors. If the setup is wrong, even perfect arithmetic produces the wrong result.

Ask: Which of those three mistakes feels easiest to make?

Expected response: Any reasonable response; use their answer to emphasize the relevant fix.

If correct: Good. Name the exact prevention rule for the mistake they mention.

If not correct: If they say none, ask them to explain why each correct setup is forced by the algebra.

Emphasize: Focus on prevention rather than blame.

Transition: Let us check the sign error first.

Page 42 — Checkpoint: wrong sign

Purpose: Diagnose the $x+3$ synthetic-division trap.

Teacher script: "A student uses $+3$ for the divisor $x+3$. What should we change?"

Ask: What should replace $+3$?

Expected response: -3 .

If correct: Correct. Solve $x+3=0$.

If not correct: Ask them to demonstrate the zero calculation.

Emphasize: The value outside synthetic division is the zero of the divisor.

Transition: Now check the zero-coefficient issue.

Page 43 — Checkpoint: missing powers

Purpose: Diagnose a missing-coefficient error.

Teacher script: "For $x^4 - 16x^3 + 4x^2 + 10x - 11$, what coefficient row is correct?"

Ask: Which row?

Expected response: 1, -16, 4, 10, -11.

If correct: Correct. There are no missing powers in this polynomial.

If not correct: Important: do not insert a zero where a power is actually present.

Emphasize: Zeros are added only for genuinely missing powers.

Transition: Let us use degree itself as another check.

Page 44 — A deeper question: why the degree drops

Purpose: Make quotient degree predictable before computation.

Teacher script: "There is a quick degree check we can use before or after a division. If a degree-5 polynomial is divided by a degree-1 polynomial, the quotient must have degree 4. In general, when the division works in the usual polynomial sense, the quotient degree is dividend degree minus divisor degree. This helps us read synthetic rows correctly.

Ask: If the dividend has degree 4 and the divisor is degree 1, what degree should the quotient have?

Expected response: 3.

If correct: Yes. That means the first bottom-row number is the coefficient of x^3 .

If not correct: If they say 4, ask which power is removed by dividing by a linear factor.

Emphasize: Degree is a useful self-check.

Transition: Let us use that as a quick checkpoint.

Page 45 — Checkpoint: quotient degree

Purpose: Check degree reasoning.

Teacher script: "A degree-5 polynomial is divided by a degree-1 polynomial. What is the degree of the quotient?"

Ask: What degree?

Expected response: 4.

If correct: Correct.

If not correct: Ask them to subtract 1 from 5.

Emphasize: Use degree as a quick sanity check before reading a synthetic row.

Transition: Now connect zero remainder to the next theorem without turning this into the next lesson.

Page 46 — The remainder connects the theorems

Purpose: Build the bridge from Remainder Theorem to Factor Theorem.

Teacher script: "Here is the important connection. For $x-b$, the Remainder Theorem says the remainder is $P(b)$. If $P(b)$ is zero, then the remainder is zero. That means the division is exact, so $x-b$ is a factor. This is the doorway to the Factor Theorem. We are only making the connection here; the full factoring work comes next.

Ask: If $P(b)=0$, what happens to the division?

Expected response: The remainder is 0, so $x-b$ is a factor.

If correct: Exactly.

If not correct: Ask them to connect $P(b)=0$ to the division identity.

Emphasize: Keep this as a bridge, not a full factoring lesson.

Transition: Let us test that statement.

Page 47 — Checkpoint: zero remainder

Purpose: Check the Factor Theorem connection.

Teacher script: "If $P(4)=0$, what can you conclude about $x-4$?"

Ask: What can you conclude?

Expected response: $x-4$ is a factor.

If correct: Correct.

If not correct: Ask them why: because the remainder when dividing by $x-4$ is $P(4)=0$.

Emphasize: The word "factor" should be justified by the zero remainder.

Transition: We have one independent synthetic problem left.

Page 48 — Practice: synthetic division with a zero

Purpose: Let the student perform a complete source-style synthetic division independently.

Teacher script: "Now you do the work. Divide $x^3 - 7x - 6$ by $x - 3$ using synthetic division. Before you write numbers, what coefficient is missing? Good— x^2 , so use 0. Set up $3 \mid 1, 0, -7, -6$. Let the student finish without help. The source answer is $x^3 - 7x - 6 = (x-3)(x^2 + 3x + 2) + 0$.

Ask: What is the complete result?

Expected response: $Q(x)=x^2+3x+2$ and $R=0$.

If correct: Good. Now ask them to turn the row into the division statement.

If not correct: If stuck, give only the next multiply/add operation rather than the answer.

Emphasize: This is the point where you want the student doing most of the work.

Transition: Finish with the general division identity.

Page 49 — Final mixed check: full division

Purpose: End with the core identity that unifies every division method.

Teacher script: "Final check. Suppose $P(x)$ is divided by $D(x)$, giving quotient $Q(x)$ and remainder R . Which statement must always be true? This is the structure behind everything we have done today.

Ask: Which statement is correct?

Expected response: $P(x) = D(x)Q(x) + R$.

If correct: Exactly. That one line is the definition of the division relationship.

If not correct: Ask the student to explain what each symbol means rather than just naming the option.

Emphasize: If they choose $P=Q+R$, bring them back to the numerical example from page 2.

Transition: This is the final conceptual check before we close. Let us finish by summarizing what the student should now be able to do.

Page 50 — Your work today

Purpose: Close the lesson, summarize the tools, and decide what needs review.

Teacher script: "Let us finish by looking back at the tools. With long division, you can divide polynomials step by step and write the quotient and remainder. With the Remainder Theorem, you can find a remainder quickly by using the zero of the divisor. With synthetic division, you can compress the arithmetic when the divisor fits the required linear form, including careful handling of missing coefficients and signs. The last question I care about is not "Which procedure do you remember?" but "Which tool fits the problem in front of you?"

Ask: Ask the student to explain, in their own words, when they would use each of the three tools.

Expected response: Remainder-only $x-b \rightarrow$ Remainder Theorem; full quotient $x-b \rightarrow$ synthetic division is a natural choice; non-monic or more general divisors $\rightarrow$ long division is often easier; always set up powers/coefficients carefully and verify when appropriate.

If correct: Good. Ask one final question: "Which part should we review before moving to the next lesson?"

If not correct: If they cannot explain the differences, revisit pages 22, 33, 34, and 35 rather than adding new material.

Emphasize: Do not use a score to end the lesson. End by stating what to review next, then let the student close the lesson. This is the final conceptual check before the next lesson.

Teacher Quick Reference

Core sequence: Divide → Multiply → Subtract → Repeat; stop when remainder degree is less than divisor degree.

Remainder Theorem: For $x-b$, remainder = $P(b)$. For $ax-b$, remainder = $P(b/a)$.

Synthetic setup: Use the zero of the divisor; list coefficients in descending powers; insert 0 for missing powers.

Method choice: Remainder only → Remainder Theorem. Full quotient by $x-b$ → synthetic division is efficient. Non-monic linear divisor → long division is often easier, though synthetic is possible. Quadratic divisor → long division.

Verification: $P(x) = D(x)Q(x) + R$.

Watch for: wrong divisor sign, missing zero coefficient, wrong quotient degree, lost remainder, incorrect non-monic adjustment, and failure to verify.

Source alignment: This script follows the finalized 50-page student HTML and the supplied MHF4U lesson package/workbook sequence. The source explicitly emphasizes long division, Remainder Theorem, synthetic division, missing coefficients, non-monic divisors, quotient form, and checking statements. See the supplied lesson package and workbook.